

ALYSSA HOPPER

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EDUCATION

B.A. in Mathematics, Minor in Neuroscience

Sept 2020 – June 2024

Carleton College

Northfield, MN

Thesis: *The Continued Fraction Expansion of e*

- Independently researched convergence properties of Euler's number, reconstructing proofs and formalizing gaps in existing literature.
- Completed hands-on molecular biology and neuroscience labs: DNA extraction, PCR, microscopy, and experiments with *Drosophila* and rodent models.
- Applied coding and data analysis in both biology and math contexts.

RESEARCH EXPERIENCE

Research Intern

Sept 2026 – Present

Dysautonomia Center, NYU Langone Health

New York, NY

- Work with the center's REDCap clinical research data under Dr. Horacio Kaufmann, supporting research on autonomic disorders including familial dysautonomia and synucleinopathies.

OTHER EXPERIENCE

Math Tutor

Sept 2018 – Mar 2026

- Provided paid one-on-one instruction in Algebra, Geometry, Pre-Calculus, Number Theory, and Statistics for students from middle school through college.
- Tutored a student who spoke only Russian, using my fluency to translate math concepts and assignments while supporting her transition to English-based learning.

Bookseller

Aug 2025 – Jan 2026

Barnes & Noble

St. Paul, MN

- Assisted customers with book selection and recommendations across genres and reading levels.
- Managed point-of-sale transactions, returns, and exchanges with attention to detail.

SKILLS

Computational

Python, MATLAB, C++, Java, HTML/CSS/JavaScript, $\text{\LaTeX}$, Excel

Analytical

Data analysis, statistical analysis, mathematical reasoning

Laboratory

DNA extraction, PCR, microscopy, *Drosophila* and rodent handling

Languages

English (native), Russian (fluent), French (intermediate)

RELEVANT COURSEWORK

Neuroscience & Biology

Developmental Neurobiology; Neurons, Circuits, and Behavior; Genetics; Sensation and Perception

Computer Science

Data Structures; Algorithms; Computation and Complexity; Artificial Life and Digital Evolution

Mathematics

Elementary Number Theory; Analytic Number Theory; Abstract Algebra; Ordinary Differential Equations